

10-min version

[intro]

- ☐ Hello everyone, my name is Zihan Bai, I'm a first year phd student working with SM and DF at NYU. Today i'll talk about how EM representations can rescue WM via pattern separation, pattern completion, and predictive recall

[sock problem]

- ☐ Picture this morning: you're getting dressed for MAMM, and you're doing a sock hunt, trying to match pairs, (assuming, we all remembered to wear socks today). Your WM holds the sock you're currently comparing. But if you remember last year's MAMM outfit where you directly compared a star sock with its dotted counterpart. Then today you know exactly what detail to look for
- ☐ This way, EM can supply that diagnostic detail, helping WM discriminate between similar ones

[mechanisms of EM-WM]

- ☐ There is already evidence that EM contributes to WM performance, but WM studies often treat EM as a confound to control. Here we aim to understand the mechanism by which EM supports WM.
- ☐ We begin by considering pattern separation at encoding, which creates distinct representations of similar items
- ☐ At retrieval, we consider pattern completion, which fills missing information from partial cues
- ☐ Importantly, predictive recall, where relevant information is reinstated in advance.

[task]

- ☐ To test this, we developed a paradigm that embeds mnemonic discrimination test within a WM n-back task, we call it the MST-Back Task, with eye-tracking.
- ☐ First, we have a 1-back encoding task. I'll show you a sequence of items. Compare each item to the previous one: clap once if it's similar but not identical; snap once if it's exactly the same; and do nothing if it's different.
- ☐ Great! Just a note that during the actual task participants saw only one item at a time without the previous, so they need to hold items in WM. I'm showing the full sequence for time purposes.
- ☐ In some trials, similar pairs, A1 and B1 appear back-to-back in the compared condition, so people have a chance to encode what differs between A and B. And in others, they are separated in time in the isolated condition, so people still encode each item but are less likely to detect the difference. We also add filler trials so each participant gets a balanced number of responses.
- ☐ Next, we move to the 2-back retrieval task. Now, compare each item to the one two trials ago. Same rules, clap for similar; snap for same; nothing for different
- ☐ Here, A B pairs were tested in 2-back as A2 and B2. Some pairs, as you may have already noticed, were previously compared, some were previously isolated, and other pairs were novel, which was a new condition we introduced. Importantly, A2 always appears, but participants have no way of knowing whether, two trials later, a B2 will appear. For example, this item has been previously compared, but the same item appeared in 2 trials. In fact, any item can intervene, including another A2. This is important because episodic memory can't help by filtering which items are relevant. It can only help by supplying additional content.
- ☐ We therefore have three conditions in 2-back: compared, isolated, and novel. This lets us compare how well people discriminate similar items with or without prior episodic memory.
- ☐ Finally, we end with an old/new recognition test on old items from 1-back. This gives us a measure of long-term memory and lets us confirm that EM was created.

[behavioral evidence for EM benefits]

- ☐ We first asked whether EM enhances WM discrimination. In 2-back, we measure accuracy using a discrimination index, and response time across three conditions. First, we see performance in the novel condition, no EM benefit. In isolated condition, people are more accurate and faster. In

compared condition, people are most accurate and fastest. Bayes factor, BF10 shows extreme evidence for discrimination index benefit, and very strong evidence for RT benefit.

- ☐ Now, I want to highlight a different type of trial: same-item identification. In these trials, the task is simply to recognize when the exact same item repeats. Here, I've highlighted the same trials in the sequence. As you can see, the pattern of responses looks similar across conditions. In fact, when we look at Bayes factors—shown here—there is strong evidence for no effect. In short, EM selectively enhances working memory discrimination

[indiv diff]

- ☐ At the individual level, we found that EM benefits draw on both memory systems.
- ☐ The left panel shows how WM same detection d' in 2-back correlates with EM benefit, which we defined as the gain in discrimination index from the compared and isolated conditions versus novel. We found greater EM benefit for subjects with better WM. The right panel shows how LTM d' from the post-experiment recognition test correlates with EM benefit, and again, we found greater EM benefit for subjects with better LTM.

[pupil]

- ☐ Next, we expected pupil dilation to reflect this benefit, since prior literature links it to episodic retrieval
- ☐ Here, we plot average pupil size change across conditions, time-locked to the similar trials in 2-back, where we saw the benefit. We observed a significant cluster: discriminating previously compared items led to pupil dilation than isolated or novel ones. Importantly, only in the compared condition did correct trials show this dilation. This means pupil dilation tracks correct discrimination of previously compared items.
- ☐ Consistent with behavior, although not shown, same-item identification didn't drive this effect, giving us online evidence of EM at the moment of successful discrimination.

[gaze sim]

- ☐ We next asked how EM is retrieved. We used gaze pattern similarity, measuring how two similar fixation patterns are, for example, A2 and B1, as a readout of what's being mentally represented.
- ☐ Before testing specific hypotheses, we compared all possible pairs of gaze similarities to a null distribution. We found all pairs are significantly higher than baseline, including same items within and across 1-back and 2-back, and similar-items within and across. Thus, we confirmed that gaze similarity tracks image similarity.

[pattern separation]

- ☐ Next using gaze similarity, we tested whether participants formed distinct episodic representations in 1-back that supported later discrimination in 2-back.
- ☐ We measured how differently they viewed A and B in 1-back. Lower A1-B1 similarity suggests they focused on more diagnostic features.
- ☐ In the compared condition, trials that were subsequently discriminated correctly had lower A1-B1 similarity. In contrast, that difference was not observed in the isolated condition.
- ☐ This provides a gaze-level signature of pattern separation. Pairs encoded with more discriminative fixations in 1-back were better discriminated in 2-back

[pattern completion]

- ☐ Next, at the moment of discrimination, we asked whether the encoded pattern was completed. We measured B2-B1 similarity as a readout of pattern completion, which is how much the gaze pattern for B reappears in 2-back relative to when it was first encoded in 1-back.
- ☐ In the compared condition, B2-B1 similarity is higher on correct trials, whereas in the isolated condition, we did not see this effect. Thus, this gaze pattern completion signature supports that: the diagnostic representation was more fully expressed during successful discrimination.

[predictive recall]

- ☐ Importantly, we tested the hypothesis that EM supports WM discrimination by predictively recalling diagnostic B representation before B appears in 2-back.
- ☐ A2 is the current item in 2-back people are viewing, and we're comparing it to B1, which is a previously encoded pairmate. In the compared condition, correct trials show higher A2-B1 similarity. And when we looked at how predictive recall unfolds across times, we found that this effect grows with more fixations
- ☐ This means when viewing A2, participants were already predictively recalling B (which is, again, its previously encoded pairmate), guiding successful upcoming discrimination.

[summary]

- ☐ To summarize. First we found evidence that EM benefits WM mnemonic discrimination, as shown by better accuracy and faster responses. At the individual level, we found that those with stronger WM and LTM benefit more from their interaction. Pupillometry showed pupil dilation tracked EM benefit, correct discrimination of previously compared items was linked to pupil dilation. Importantly, gaze pattern reveals the mechanism: Diagnostic gaze patterns were encoded via pattern separation, and were predictively recalled to prepare for upcoming discrimination, and were more fully completed at the moment of successful discrimination.
- ☐ Together, these findings show how EM rescues WM in continuous tasks.